

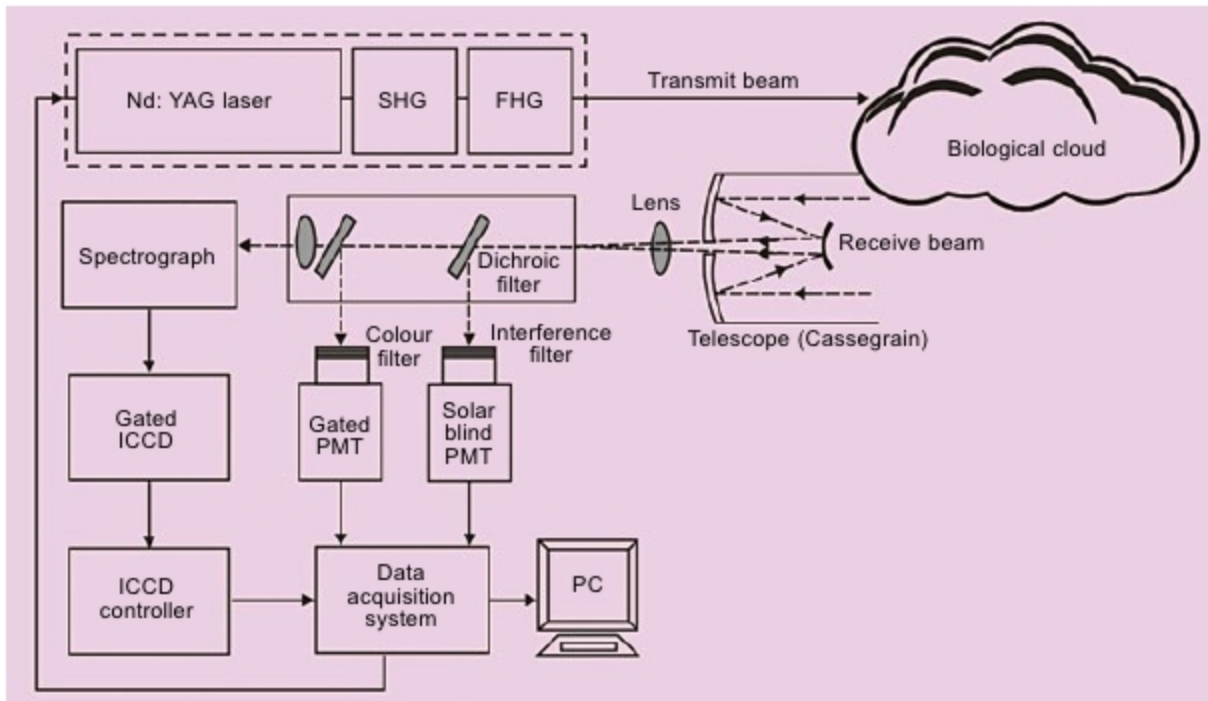


Fig. 5: UV LIF lidar system for detection of biological agents



Fig. 6: UV LIF lidar of Norwegian Defence Research Establishment

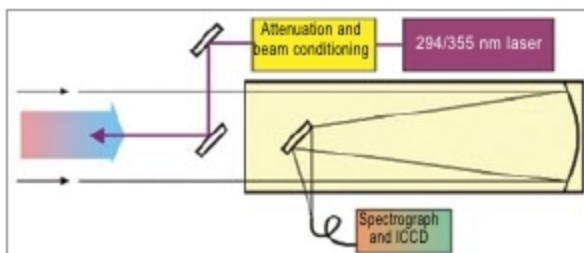


Fig. 7: Simplified block schematic of Norwegian biological lidar

digitised and fed to the data processor to extract the desired information on the type and concentration of chemical species.

Most information on lidar complexes specifically designed for detection and identification of chemical warfare agents is classified and not available on public domains. There are many lidar systems available for monitoring atmospheric pollutants and analysing meteorological parameters. There are some systems that are also capable of stand-off detection of chemical warfare agents such as organophosphates.

One such mobile lidar complex from Laser Systems, Russia is shown



Fig. 8: Long-range biological stand-off detection system

in Fig. 4. The lidar complex combines in itself aerosol lidar, wind lidar, long wave DIAL lidar and short-wave DIAL lidar, with maximum detection ranges of 20km, 5km, 5km and 3km, respectively. The system can detect hydrocarbons, Freon gases, organophosphates and so on. The scanning lidar offers azimuth field-of-view of $\pm 180^\circ$ and elevation field-of-view of -15° to $+90^\circ$.

Detection of biological warfare agents

UV LIF lidar. There were several factors during 1995 - 2005 that led to increased international concern about biological warfare agents. This was the result of coordinated attacks in Tokyo using sarin gas in 1995, suspicions of an Iraqi biological warfare programme, and anthrax attacks on US Congress members and media outlets in 2001. Devastating effects of biological warfare and the ease with which a biological agent attack can be

executed was amply evident from 2001 anthrax attacks in the US.

While the technologies needed for detection and identification of chemical warfare agents are more mature, stand-off detection of biological warfare agents and their discrimination from background aerosols is extremely challenging. This is primarily due to the distinction between innocuous ambient bacteria and virulent microbes, which amounts to subtle differences in their molecular make up.

As these subtle differences involve a very small percentage of molecules, these produce a very small change in their optical signatures. This makes it extremely difficult to reliably detect and discriminate between these potentially-dangerous molecules.

Processes of detection and discrimination are further complicated due to constant variation in growth media. Contaminants associated with the processing of biological warfare agents affect their optical signatures.

Stand-off detection of biological warfare agents is based on LIF effect. It is the emission from atoms or molecules after these have been excited to go to higher energy levels by excitation by another laser. Emission takes place at a wavelength that is higher than the wavelength of the laser light exciting it.

Biological warfare agent molecules mainly constitute aromatic amino acids and coenzymes. Aromatic amino acids such as tryptophan, tyrosine and phenylalanine absorb laser radiation at 280nm - 290nm and fluoresce in 300nm - 400nm bands. It is therefore possible to detect biological warfare agents by using excitation by UV laser of suitable wavelength. Also, discrimination of biological agents can be achieved only from LIF signal, because fluorescence cross-section for particles in the range of 1 μ m to 10 μ m size are sufficiently large to make single particle interrogation feasible.

Fig. 5 shows the block schematic arrangement of a typical monostatic UV LIF lidar system. Principle components of this system include frequency tripled or quadrupled Nd: YAG laser [respectively emitting at 355nm (frequency tripled output) and 266nm (frequency quadrupled output)], telescope for both transmission and